

Date	15 March 2026
Team ID	xxxxxx
Project Name	Credit Card Approval Prediction
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should do).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should be).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Bank agents and administrators must securely log into the system using Multi-Factor Authentication (MFA) or SSO (Single Sign-On).	High
2	Authorization levels	System must support role-based access control (RBAC). Admins can retrain the ML model and configure thresholds; Agents can only submit application data and view prediction results.	High
3	External interfaces	The system must connect via API to external credit bureaus (e.g., Equifax, Experian) to fetch baseline credit scores if available.	Medium
4	Transactions processing	The system must route incoming application data through an automated preprocessing pipeline and the ML classification model to output an approval decision.	High

5	Reporting	The dashboard must generate real-time XAI (Explainable AI) reports using SHAP values for individual applicants, and a monthly aggregate report of approval rates.	High
S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
6	Business rules, process etc.	If the ML model predicts an approval probability between 45% and 55%, the application must be automatically flagged for manual review rather than rejected.	High
7	Compliance to laws or regulations	The data handling processes must comply with GDPR and CCPA to ensure applicant financial data is processed securely and with consent.	High
8	Other	Automated data drift monitoring must alert the admin if incoming data distributions change significantly from the training data.	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The ML model prediction and generation of the explainability (SHAP) visual report must execute quickly so the bank agent doesn't wait.	Response time < 3 seconds per application.
2	Scalability	The backend infrastructure must scale to handle spikes in application volume, especially during credit card promotional periods.	Supports up to 1,000 concurrent prediction requests.
3	Security & Data Privacy	All sensitive applicant information (income, debt, SSN/ID) must be encrypted both in transit and at rest.	End-to-end AES-256 encryption and TLS 1.3 protocol.
4	Reliability & Availability	The prediction service API must be highly available for bank branches across different time zones.	99.99% uptime per month.
5	Usability & Accessibility	The frontend dashboard must be intuitive for non-technical banking staff and compliant with accessibility standards.	Complies with WCAG 2.1 AA standards.
6	Other: Explainability Standard	Every automated rejection must be accompanied by a clear, human-readable justification to prevent "black box" decisions.	100% of ML predictions accompanied by a mapped XAI report.